

EXPERIMENT NO. 7**AIM: Traffic Data Analysis using Wireshark.****Exercise**

1. Turn on wireshark.
2. Then select your respective network adapter (eg. Ethernet) and start() capturing the data.
3. Now, go to your browser.
4. Type **172.16.12.25** in your search engine.
5. After 2 seconds go to wireshark and stop() capturing data.
6. Go to the wireshark file menu and click on save.
7. Save your file as pr-7_xyz_sample1.
8. Repeat steps 1,2,3.
9. Type **172.16.0.1:8090** in your search engine.
10. After 2 seconds go to wireshark and stop() capturing data.
11. Go to the wireshark file menu and click on save.
12. Save your file as pr-7_xyz_sample2.

(where xyz is last three digits of your enrolment number.)

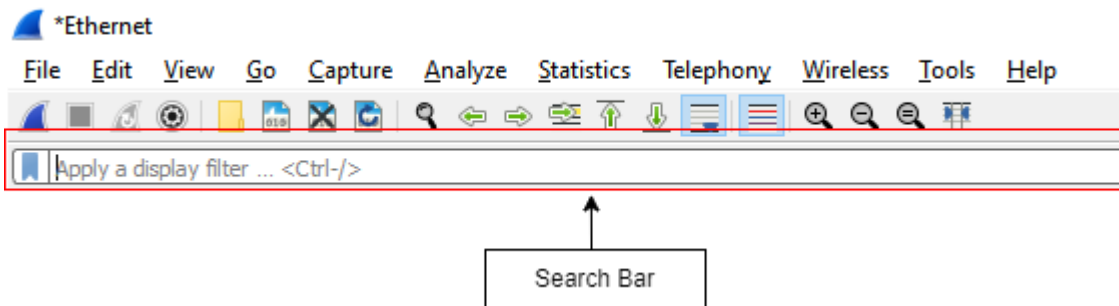
Based on this above steps answer the following questions and fill the below tables with respect to fields.

Observe the columns and write its purpose in the table below.

Apply a display filter ... <Ctrl-/>						
No.	Time	Source	Destination	Protocol	Length	Info
Fields	Description					
No.	It is the number of total captured pkts					
Time	It is the time when pkt is captured					
Source	Where the Source address is displayed					
Destination	Where the Destination address is displayed					
Protocol	Where the used protocol is displayed					
Length	Where the length of pkt is mentioned					
Info	Where the brief info about pkt is given					

Observe the capture data and write description as per following table for sample 1.

Fields	Description/Answer
Time difference between packet no. 1 and packet no. 10	0.019867
Time difference between packet no. 15 and packet no. 40	0.031957
Source of packet no. 2	e8:d8:d1:43:6e:42
Destination of packet no.2	33:33:ff:70:76:a0
Transport layer protocol of packet no. 2	ICMPv6
Network layer protocol of packet no. 2	IPv6
Data link layer(Layer 2) protocol of packet no. 1	Ethernet



Use Different Filters and write the description in the table below for sample 1.

***Ethernet**

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

ip.dst==172.16.12.25

No.	Time	Source	Destination	Protocol	Length	Info

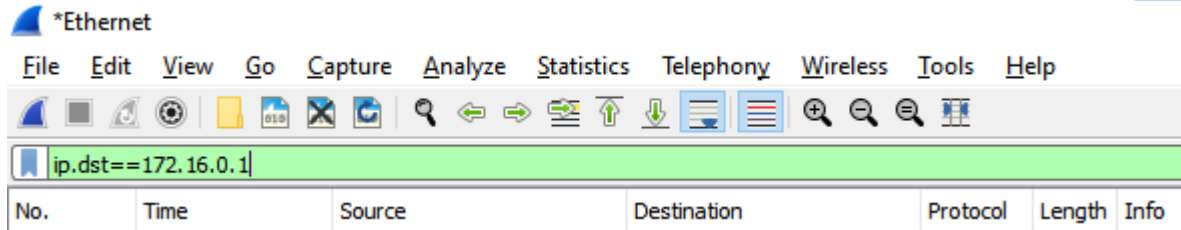
Fields	Description/Answer
First packet No. sent by your PC(Source). (Query : ip.src==192.168.68.21)	9
First packet received from destination (Query : ip.dst==172.16.12.25).	487475
Time of packet sent by your PC (Query : ip.src==192.168.68.21).	237.592905

Time of packet received from destination (Query : ip.dst==172.16.12.25)	0.019867
Total number of ARP packets (Query : arp)	105282
Total number of TCP packets (Query : tcp)	52367
Total number of HTTP/HTTPS packets (Query : http https)	20863
Time at which first TCP packet sent (Query : tcp)	0.040565
Time at which HTTP request sent (Query : http)	0.484652
Time at which HTTP reply received (Query : http)	1.142554
Time difference between first TCP packet and first HTTP packet (Query : http tcp)	0.446265
Time difference between http request and http reply (Query : http)	0.894684

observe the capture data and write description as per following table for sample 2.

Fields	Description/Answer
Time difference between packet no. 1 and packet no. 10	0.022365
Time difference between packet no. 15 and packet no. 40	0.066826
Source of packet no. 1	a6:d8:h1:33:8e:72
Destination of packet no.1	b3:d6:ff:60:76:d0
Transport layer protocol of packet no. 1	TCP
Network layer protocol of packet no. 1	IPv4
Data link layer(Layer 2) protocol of packet no. 1	Ethernet

Use Different Filters and write the description in the table below for sample 2.



Fields	Description/Answer
First packet No. sent by your PC(Source).	12
First packet received from destination.	38967
Time of packet sent by your PC	225.645517
Time of packet received from destination	0.078865
Total number of ARP packets	182582
Total number of TCP packets	95667
Total number of HTTP/HTTPS packets	26563
Time at which first TCP packet sent	0.087755
Time at which HTTP request sent	0.654652
Time at which HTTP reply received	1.851654
Time difference between first TCP packet and first HTTP packet	0.325465
Time difference between http request and http reply	0.231894

CONCLUSIONS:

Post-Lab Assignment Questions
(Write answers in file pages, do not take print of this page)

1. A user is unable to ping a system on the network. How can wireshark be used to solve the problem?
[Hint : ping any machine then go to wireshark and observe the screen]
2. Which wireshark filter can be used to monitor outgoing packets from a specific system on the network.
3. Which query should you use to filter all web traffic?
4. A user raises a ticket stating that he is unable to access any websites, but is able to ping any IP address on the internet. How would you use wireshark to identify the problem?
[Hint : type query as dns and observe the dns packets and write description as you understood.]
5. Is it possible to find out if any packet was lost while capturing data with Wireshark? If yes, then how?
[Hint : go to wireshark statistics>summary and look at the parameters.]
6. Which query would you search to remove all packets that were captured before and after a specific time range?
7. List out examples where wireshark can be used.
8. Why do you think it's important to monitor network traffic on enterprise systems?
9. Which wireshark filter(query) can be used to check all incoming requests to a HTTP Web server